

Production Codec Reference Guide

VideoWise Codec Compatibility Reference

For production environments without Python access

Quick Reference: Use this document to verify codec compatibility before shows, train operators, and troubleshoot playback issues without running VideoWise.

Document Purpose

This reference consolidates codec requirements from VideoWise's 23 system checkers into a single document for:

- Live production operators checking codec compatibility before showtime
- Training teams on codec requirements for different systems
- Troubleshooting playback issues during production
- Offline reference when Python tools are unavailable

Table of Contents

1. [Codec Quick Reference](#)
 2. [Live Production Systems](#)
 3. [Church & Theatre Presentation](#)
 4. [Media Players & VJ Software](#)
 5. [Browsers](#)
 6. [Social Media Platforms](#)
 7. [Common Codec Types Explained](#)
 8. [FFmpeg Quick Commands](#)
-

Codec Quick Reference

System Type	Best Codec	Alternative	Container
Live Production	HAP, ProRes	H.264, DNxHD	MOV
Church/Theatre	ProRes 422, H.264	HAP	MOV, MP4
VJ Software	DXV, HAP	ProRes	MOV
Browsers	H.264	VP9, AV1	MP4
Social Media	H.264 Baseline	HEVC (limited)	MP4

Table 1: Quick codec selection guide by system type

Live Production Systems

CasparCG Server

Supported Codecs: H.264, ProRes, DNxHD, DNxHR, MPEG-2, MJPEG, HAP, NotchLC

Optimal Codecs:

- **HAP** (all variants): GPU-accelerated, best for real-time playback
 - HAP: Standard GPU acceleration
 - HAP Alpha: GPU acceleration with transparency
 - HAP Q: Highest quality HAP variant for broadcast
- **ProRes 4444**: Professional quality with alpha channel support
- **NotchLC**: High-quality real-time playback, popular in broadcast

Container Requirements:

- H.264: MP4 or MOV
- ProRes: MOV only
- DNxHD/DNxHR: MOV or MXF
- HAP: MOV only (critical requirement)
- NotchLC: MOV only

Critical Requirements:

- **Constant Frame Rate (CFR)**: Variable frame rate causes timing and sync issues in live production

- **4K Bandwidth:** Videos over 200 Mbps may stress system bandwidth
 - Recommendation: Use HAP codec for GPU-accelerated 4K playback

Common Issues:

- VFR (Variable Frame Rate) detection: Convert to CFR matching production framerate
 - Container mismatch: HAP in MP4 will not play (must be MOV)
 - High bitrate 4K: Consider HAP codec to offload decoding to GPU
-

PlayoutBee

Supported Codecs: H.264, ProRes, HAP (all variants)

Optimal Codecs:

- **HAP:** Best choice for GPU acceleration and low CPU usage
 - HAP: Standard, optimal for most uses
 - HAP Alpha: GPU acceleration with transparency for overlays
 - HAP Q: High quality (warning: demanding on Raspberry Pi)
- **ProRes:** Fully supported, professional quality
 - ProRes 4444: Alpha channel support for transparency workflows
- **H.264:** Good compatibility with hardware acceleration

Platform-Specific Guidance:

Desktop (Windows/Mac):

- HAP Q: Excellent for high-quality playout
- ProRes: Full support, professional workflows
- H.264: Hardware acceleration available

Raspberry Pi:

- HAP: Recommended (GPU-accelerated, low resource)
- HAP Q: May be too demanding (use standard HAP instead)
- H.264: Keep under 50 Mbps bitrate
- ProRes: Too demanding (high data rates exceed Pi capabilities)

- Resolution: 1080p or lower recommended

Container Requirements:

- HAP: MOV only
 - H.264/ProRes: MOV or MP4
-

vMix

Supported Codecs: All major codecs (H.264, HEVC, ProRes, DNxHD, DNxHR, MJPEG)

Optimal Codecs:

- **ProRes:** Well-supported, excellent quality
- **H.264:** Hardware acceleration available, good performance
- **DNxHD/DNxHR:** Professional codecs for broadcast workflows

Bitrate Recommendations:

- **Under 100 Mbps:** Optimal performance
- **100-200 Mbps:** Warning zone (monitor for performance issues)
- **Over 200 Mbps:** Very high risk of dropped frames
 - Recommendation: Transcode to 100-150 Mbps

Resolution Warnings:

- **4K (3840x2160):** Requires powerful hardware
 - Ensure system meets vMix's 4K requirements
 - CPU/GPU intensive, test before going live

Container: MP4, MOV, MKV all supported

OBS Studio

Supported Codecs: H.264, HEVC, AV1, VP8, VP9, ProRes, DNxHD

Recommended Codecs: H.264, HEVC, AV1 (hardware acceleration available)

Container Formats:

- **MKV/Matroska:** OBS's default recording format

- **MP4/MOV:** Good compatibility with video editing software
- **WebM:** For VP8/VP9 codecs

Notes:

- H.264/HEVC/AV1 may have hardware decoding available
 - OBS is very flexible with codec support through FFmpeg
-

QLab

Recommended Codecs (in order of preference):

1. **ProRes Proxy:** Best performance, optimized for playback
2. **ProRes LT:** Excellent performance, good quality
3. **ProRes 422:** Professional quality, reliable playback
4. **ProRes 4444:** For videos requiring alpha channel (transparency)

Not Recommended:

- **H.264:** Poor performance when scrubbing or changing playback speed
 - Not optimized for variable-speed playback
 - Recommendation: Convert to ProRes 422 Proxy or LT

Container: MOV or MP4 (MOV preferred for ProRes)

Why ProRes?

- QLab is optimized for frame-accurate scrubbing
 - ProRes variants are intra-frame codecs (every frame independent)
 - H.264 uses inter-frame compression (frames depend on other frames)
 - Scrubbing H.264 requires decoding multiple frames = poor performance
-

ProPresenter

Supported Codecs: H.264, HEVC, ProRes, ProRes 4444, HAP

Optimal Codecs:

- **HAP:** Best performance (GPU-accelerated real-time playback)
- **ProRes:** Fully supported, professional quality
 - ProRes 4444: Alpha channel support
- **H.264:** Compatible, good for standard playback
- **HEVC:** Supported

Container: MOV or MP4 (both work well)

Notes:

- HAP provides best performance for real-time presentation
 - ProRes 4444 required for transparency/alpha channel
-

Church & Theatre Presentation

Wirecast

Supported Codecs: H.264, HEVC, ProRes, DNxHD, DNxHR, MJPEG

Recommended Codecs:

- **H.264:** Fully supported, hardware acceleration available
- **ProRes:** Well-supported, professional codec

Container: MP4 or MOV preferred (WMV/AVI supported but not recommended)

Bitrate Limit: Keep under 150 Mbps for smooth operation

- High bitrate (>150 Mbps) can strain CPU during encoding
-

Playback Pro

Recommended Codecs:

- **ProRes 422:** Optimal balance of quality and performance for theatre
- **H.264:** Supported with specific bitrate requirements

Critical Requirement: MOV container only

- Playback Pro only accepts .MOV files

- Will not load MP4, AVI, or other containers

H.264 Bitrate Recommendations:

HD (1920x1080):

- Recommended: 15-30 Mbps VBR (Variable Bitrate)
- Below 15 Mbps: May show quality issues
- Above 30 Mbps: Wasted bitrate

4K/UHD (3840x2160):

- Recommended: 30-40 Mbps VBR
- Below 30 Mbps: Quality degradation
- Above 40 Mbps: Diminishing returns

Remux to MOV:

```
ffmpeg -i input.mp4 -c copy output.mov
```

EasyWorship

Platform: Windows only

Native Support (No Codec Packs Needed):

- **H.264 in MP4/MOV/M4V:** Works out of the box in EasyWorship 7+

Windows Media Codecs (Built into Windows):

- WMV (Windows Media Video)
- MPEG-2

Other Codecs: May require additional codec packs

- Recommendation: Convert to H.264 in MP4 for guaranteed compatibility

Container: MP4, MOV, M4V (native support), WMV, AVI (Windows support)

Media Players & VJ Software

VLC Media Player

Codec Support: Universal (plays virtually everything via FFmpeg)

Hardware Acceleration Available For:

- H.264
- HEVC (H.265)
- VP9
- AV1

Enable Hardware Decoding: VLC Preferences → Input/Codecs → Hardware-accelerated decoding

Performance Notes:

- **Very High Bitrate** (>300 Mbps): May cause stuttering
 - Requires fast storage (NVMe SSD)
- **8K Video** (7680x4320): Requires powerful CPU/GPU and fast storage
 - Use VLC 3.0+ with hardware decoding enabled

Container: All major containers fully supported

Resolume Arena/Avenue

Optimal Codecs (GPU-accelerated):

- **DXV** (DXV, DXV2, DXV3): Resolume's proprietary GPU codec, best performance
- **HAP** (HAP, HAP Alpha, HAP Q, HAP Q Alpha): GPU-accelerated, cross-platform
 - HAP Alpha: Perfect for overlays and VJ graphics with transparency
 - HAP Q: Highest quality HAP, performance equal to DXV

Supported but Not Optimal:

- **H.264:** CPU-based decoding, not GPU
 - Warning: Performs poorly with multiple layers

- Recommendation: Convert to DXV or HAP
- **ProRes:** CPU-based (Mac has better support than Windows)
 - Mac: Works but limits layer count
 - Windows: Slow, not optimized
- **HEVC:** CPU-intensive, not optimized for VJ work

Container Requirements:

- HAP: MOV only
- DXV: MOV recommended

Layer Count Factors:

- **4K Video:** Limits simultaneous layers (4x bandwidth of 1080p)
 - Recommendation: Use 1080p for more layers, or DXV/HAP for 4K
- **High Bitrate** (>200 Mbps): Stresses disk I/O even with GPU codecs
 - Recommendation: Use DXV or HAP with moderate bitrate

Platform-Specific:

- **Windows:** DXV or HAP only (avoid ProRes)
- **Mac:** DXV, HAP preferred (ProRes acceptable but CPU-limited)

Mitti

Platform: Mac only

Recommended Codecs (Mitti transcodes everything to these):

- **ProRes:** Hardware accelerated on Apple Silicon (M1/M2/M3)
- **HAP:** GPU-accelerated, especially for 4K and multi-output
 - HAP Alpha: Optimal for transparency

Transcoding Strategy:

- Mitti can transcode any codec to ProRes or HAP
- Upload any format, use built-in transcoding feature
- **ProRes:** Best for single output (SDI)
- **HAP:** Best for multi-output (HDMI/DisplayPort via GPU)

Container: MOV preferred (MP4 works but MOV is better)

4K Guidance:

- **Single Output:** ProRes on Apple Silicon performs excellently
 - **Multi-Output:** HAP for GPU path (external displays)
 - **High Bitrate** (>250 Mbps): May cause dropped frames
 - Recommendation: ProRes 422 or HAP with moderate bitrate
-

Millumin

Platform: Mac only

Codec Support: All QuickTime formats (uses QuickTime and AVFoundation)

Recommended Codecs:

- **ProRes:** Excellent choice, native Mac codec with Apple Silicon hardware acceleration
- **HAP:** Optimal for projection mapping (GPU-accelerated)
 - Ideal for multi-projector setups
- **H.264:** Works but not recommended for projection
 - CPU-based, can limit real-time performance
 - Recommendation: Use ProRes or HAP for projection mapping

Container: MOV preferred (QuickTime native to macOS), MP4 supported

4K Projection: Requires powerful Mac, use HAP codec for best performance

ProVideoPlayer (PVP)

Supported Codecs: DXV, HAP, ProRes, H.264, HEVC

Optimal Codecs:

- **DXV:** Works perfectly with PVP timecode features
 - GPU-accelerated with excellent timecode sync

- **HAP:** Excellent performance for multi-layer playback
 - HAP Alpha: GPU-accelerated overlays with transparency
- **ProRes:** Fully supported, professional quality
 - ProRes 4444: Alpha channel support

Also Supported:

- H.264: Hardware acceleration available (consider DXV/HAP for multi-layer)
- HEVC: Hardware acceleration available

Container: MOV or MP4

Timecode Support: DXV and HAP work excellently with PVP's timecode triggering and following features

Browsers

Safari

Supported Codecs: H.264, HEVC (H.265) **only**

Container: MP4 (works best)

Incompatible Codecs: VP8, VP9, AV1, ProRes, DNxHD (all unsupported)

Critical Note: Safari has the most limited codec support of all major browsers

- Recommendation: Always use H.264 in MP4 for maximum browser compatibility
-

Chrome

Supported Codecs: H.264, VP8, VP9, AV1

Hardware Acceleration: Available for H.264, VP9, AV1

Container: MP4, WebM

Notes:

- VP9 in WebM: Native Google format
 - H.264 in MP4: Universal compatibility
-

Firefox

Supported Codecs: H.264, VP8, VP9, AV1

Partially Supported:

- **HEVC:** Limited support (requires Windows 10+ with HEVC Video Extensions)

Optimal Pairings:

- VP8/VP9 in WebM: Firefox's preferred format
- H.264 in MP4: Universal browser compatibility

Recommendation: Use H.264 in MP4 or VP9 in WebM for broad compatibility

Social Media Platforms

Instagram

Codec: H.264 **only** (Instagram re-encodes everything)

H.264 Profile: Baseline (not Main or High)

- Non-Baseline profiles will be re-encoded = quality loss
- Command: `ffmpeg -profile:v baseline`

Resolution: Maximum 1080x1920 (vertical video)

- Higher resolutions will be downscaled

File Size: 100MB maximum for feed posts

Duration: 60 seconds (feed), 90 seconds (reels)

Twitter/X

Recommended Codec: H.264 High Profile

Container: MP4 or MOV

File Size Limits:

- **Standard Accounts:** 512MB
- **Premium Accounts:** 8GB

Duration Limit: 140 seconds (standard accounts)

YouTube

Recommended Codec: H.264 High Profile with CABAC

Container: MP4 (preferred), MOV, AVI (accepted)

File Size: Maximum 256GB

Duration: Maximum 12 hours

Notes:

- YouTube re-encodes all uploads to multiple formats
 - Upload as H.264 High Profile for best quality control
 - Faster processing with MP4 container
-

TikTok

Recommended Codec: H.264 (universal compatibility across all devices)

Not Recommended:

- **HEVC:** 15-20% of US iOS devices have HEVC compatibility issues
 - Convert to H.264 for universal compatibility

Resolution: 1080x1920 optimal (9:16 aspect ratio for vertical video)

- Higher resolutions downscaled to 1080p

Bitrate Recommendations:

- **Optimal:** 8-15 Mbps
- **Below 5 Mbps:** TikTok flags as low quality
- **Above 20 Mbps:** TikTok flattens (wasted bitrate)

File Size Limits:

- **Mobile Upload:** 287MB
- **Desktop Upload:** 10GB

Container: MP4 or MOV

Vimeo

Recommended Codec: H.264 (fast upload and optimal processing)

Also Accepts: ProRes (not recommended due to large file sizes and slow uploads)

Container: MP4 or MOV

Bitrate Recommendations by Resolution:

Resolution	Recommended Bitrate
720p	5-10 Mbps
1080p	10-20 Mbps
4K	40-50 Mbps

Table 2: Vimeo bitrate recommendations

Facebook

Recommended Codec: H.264 (universal compatibility across Feed, Stories, and Ads)

Also Supports: HEVC, VP9, AV1 (for Reels)

Container: MP4 or MOV (preferred), AVI/WMV (supported but not recommended)

File Size: Maximum 4GB

Duration: Maximum 240 minutes

Resolution: 720p or higher provides good quality

Common Codec Types Explained

H.264 (AVC)

Type: Inter-frame compression (CPU-based)

Best For: Universal compatibility, browsers, social media, standard playback

Profiles:

- **Baseline:** Maximum compatibility (Instagram, older devices)
- **Main:** Good balance (most consumer devices)
- **High:** Best quality (YouTube, Twitter, desktop)

Pros: Universal support, hardware acceleration widely available, small file sizes

Cons: CPU-intensive decoding, poor scrubbing performance, not ideal for multi-layer VJ work

ProRes

Type: Intra-frame compression (every frame independent)

Best For: Theatre playback (QLab, Playback Pro), editing, archival, Mac workflows

Variants:

- **ProRes Proxy:** Lowest bitrate, optimized for playback (QLab recommended)
- **ProRes LT:** Light quality, good for most uses
- **ProRes 422:** Standard professional quality
- **ProRes 422 HQ:** Higher quality for broadcast
- **ProRes 4444:** Supports alpha channel (transparency)

Pros: Frame-accurate scrubbing, professional quality, Apple Silicon hardware acceleration

Cons: Large file sizes, MOV container only, CPU-based on Windows

HAP

Type: GPU-accelerated texture compression

Best For: VJ software (Resolume, Mitti), live production (CasparCG, PlayoutBee), multi-layer playback

Variants:

- **HAP:** Standard GPU acceleration, low CPU usage
- **HAP Alpha:** GPU acceleration with alpha channel
- **HAP Q:** High quality, larger file size
- **HAP Q Alpha:** High quality with alpha channel

Pros: GPU-accelerated (offloads CPU), excellent for multi-layer, real-time performance

Cons: MOV container only, larger than H.264, not universally supported

DXV

Type: GPU-accelerated compression (Resolume proprietary)

Best For: Resolume Arena/Avenue, ProVideoPlayer

Variants: DXV, DXV2, DXV3 (increasing quality)

Pros: Best performance in Resolume, GPU-accelerated, excellent timecode support

Cons: Proprietary to Resolume (not cross-platform like HAP), MOV container only

DNxHD/DNxHR

Type: Intra-frame professional codec (Avid)

Best For: Broadcast workflows, professional editing, Avid systems

Pros: Professional quality, supported in vMix, CasparCG, Wirecast

Cons: Large file sizes, primarily for professional workflows

FFmpeg Quick Commands

Convert to H.264 for Social Media

Instagram (Baseline Profile):

```
ffmpeg -i input.mov -c:v libx264 -profile:v baseline -level 3.0 -pix_fmt yuv420p  
-c:a aac -ar 48000 -b:a 128k -movflags +faststart output.mp4
```

YouTube/Twitter (High Profile):

```
ffmpeg -i input.mov -c:v libx264 -profile:v high -preset slow -crf 18  
-c:a aac -b:a 192k -movflags +faststart output.mp4
```

Convert to ProRes for Theatre

ProRes 422 Proxy (QLab):

```
ffmpeg -i input.mp4 -c:v prores_ks -profile:v 0 -c:a pcm_s16le  
output.mov
```

ProRes 422 (Playback Pro):

```
ffmpeg -i input.mp4 -c:v prores_ks -profile:v 2 -c:a pcm_s16le  
output.mov
```

ProRes 4444 (with alpha):

```
ffmpeg -i input.mov -c:v prores_ks -profile:v 4 -pix_fmt yuva444p10le  
-c:a pcm_s16le output.mov
```

Convert to HAP for VJ Work

HAP (standard):

```
ffmpeg -i input.mp4 -c:v hap -format hap output.mov
```

HAP Q (high quality):

```
ffmpeg -i input.mp4 -c:v hap -format hap_q output.mov
```

HAP Alpha (with transparency):

```
ffmpeg -i input.mov -c:v hap -format hap_alpha output.mov
```

Remux (Change Container Without Re-encoding)

MP4 to MOV:

```
ffmpeg -i input.mp4 -c copy output.mov
```

MOV to MP4:

```
ffmpeg -i input.mov -c copy -movflags +faststart output.mp4
```

Convert to Constant Frame Rate (CFR)

Fix VFR for Live Production:

```
ffmpeg -i input.mp4 -c:v libx264 -r 30 -c:a copy output.mp4
```

Replace 30 with your production framerate (23.976, 24, 25, 29.97, 30, 50, 59.94, 60)

Check Video Information

Get Codec, Container, Resolution, Bitrate:

```
ffprobe -v error -show_entries format=format_name,size,bit_rate  
-show_entries stream=codec_name,width,height,r_frame_rate  
-of default=noprint_wrappers=1 input.mp4
```

Troubleshooting Common Issues

Issue: "Video won't load in CasparCG"

Check:

1. Is it HAP in MOV container? (HAP requires MOV, not MP4)
2. Is it a supported codec? (H.264, ProRes, HAP, DNxHD, NotchLC)
3. Is it constant frame rate (CFR)? (VFR causes issues)

Solution: Convert to HAP in MOV or ProRes in MOV

Issue: "Dropped frames during playback in vMix"

Check: Bitrate (over 200 Mbps causes issues)

Solution: Transcode to 100-150 Mbps:

```
ffmpeg -i input.mov -c:v libx264 -b:v 120M -c:a copy output.mp4
```

Issue: "QLab scrubbing is laggy"

Check: Codec (H.264 performs poorly when scrubbing)

Solution: Convert to ProRes 422 Proxy:

```
ffmpeg -i input.mp4 -c:v prores_ks -profile:v 0 -c:a pcm_s16le  
output.mov
```

Issue: "Resolume can only play 2-3 layers"

Check: Codec (H.264 is CPU-based, not GPU)

Solution: Convert to HAP or DXV for GPU acceleration:

```
ffmpeg -i input.mp4 -c:v hap -format hap_q output.mov
```

Issue: "Instagram re-encoded my video and quality dropped"

Check: H.264 profile (Instagram requires Baseline, not Main or High)

Solution: Convert to H.264 Baseline:

```
ffmpeg -i input.mp4 -c:v libx264 -profile:v baseline -c:a copy  
output.mp4
```

Issue: "Safari won't play video but Chrome works"

Check: Codec (Safari only supports H.264 and HEVC)

Solution: Convert to H.264 in MP4:

```
ffmpeg -i input.webm -c:v libx264 -c:a aac output.mp4
```

Issue: "Playback Pro won't load video"

Check: Container (Playback Pro requires MOV)

Solution: Remux to MOV:

```
ffmpeg -i input.mp4 -c copy output.mov
```

Quick Decision Chart

I need to play video in...

Live production (CasparCG, PayoutBee, vMix):

→ Use HAP in MOV (GPU acceleration) or ProRes in MOV (quality)

Theatre (QLab, Playback Pro):

→ Use ProRes 422 Proxy in MOV (QLab) or ProRes 422 in MOV (Playback Pro)

VJ software (Resolume, Mitti):

→ Use DXV in MOV (Resolume only) or HAP in MOV (cross-platform)

Church presentation (ProPresenter, EasyWorship):

→ Use H.264 in MP4 (universal) or HAP in MOV (best performance)

Web browsers:

→ Use H.264 in MP4 (Safari compatible) or VP9 in WebM (Chrome/Firefox)

Social media:

→ Use H.264 Baseline in MP4 (Instagram) or H.264 High in MP4 (YouTube/Twitter)

Container Format Notes

MOV (QuickTime)

Required For: HAP, DXV, ProRes (best), Playback Pro (mandatory)

Best For: Mac workflows, professional codecs, theatre playback

MP4

Required For: Web browsers, social media

Best For: H.264, HEVC, universal compatibility

MKV (Matroska)

Best For: OBS Studio (default recording format)

WebM

Best For: VP8/VP9 codecs for web (Chrome/Firefox)

Alpha Channel (Transparency) Support

Codecs with Alpha Channel:

- ProRes 4444
- HAP Alpha
- HAP Q Alpha

Use Cases:

- Overlays and graphics in live production
- VJ graphics with transparency
- Multi-layer compositions

Container: MOV only for alpha channel support

Constant vs Variable Frame Rate

Constant Frame Rate (CFR): Every frame has same duration (required for live production)

Variable Frame Rate (VFR): Frame duration varies (causes timing issues in live)

Live Production Systems Require CFR:

- CasparCG
- vMix
- PlayoutBee
- ProPresenter

Convert VFR to CFR:

```
ffmpeg -i input.mp4 -c:v libx264 -r 30 -c:a copy output.mp4
```

Hardware Acceleration

Codecs with Hardware Decoding:

- H.264 (widely supported)
- HEVC (modern GPUs)
- VP9 (Chrome, Firefox)
- AV1 (newest GPUs)

GPU-Accelerated Codecs (offload CPU):

- HAP (all variants)
- DXV (Resolume proprietary)

When to Use GPU Codecs:

- Multi-layer VJ work (Resolume, Mitti)
 - Live production with multiple video sources
 - 4K playback on moderate hardware
-

Document Version

Based On: VideoWise [compatibility.py](#) (23 system checkers)

Last Updated: February 2026

Systems Covered: 23 (Live Production, Church/Theatre, VJ Software, Browsers, Social Media)

Additional Resources

VideoWise GitHub: <https://github.com/KnowOneActual/video-codec-checker>

VideoWise Documentation: See /docs folder in repository

FFmpeg Documentation: <https://ffmpeg.org/documentation.html>